Elies Amari Milliti

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PROFILE

Neuro-X MSc student at EPFL specializing in **brain-computer interfaces**, **neural signal processing**, and **machine learning** for **neural data**. Experience developing **real-time EEG decoding** pipelines, **closed-loop** neurotechnology systems, and **adaptive stimulation artifact cancellation** methods. Background in microengineering, biomedical signal processing, and experimental neuroscience.

RESEARCH & ENGINEERING EXPERIENCE

INL, EPFL | Semester Project — BCI Artifact Cancellation — *Jul. 2026 – Sep. 2026*

- Developing an **adaptive artifact cancellation pipeline** for simultaneous neural recording and stimulation.
- Implementing **PASTd-based subspace tracking** for real-time artifact suppression in multichannel EEG.
- Evaluating performance using **time-domain, spectral, and spatial signal analysis**.

EPFL AI Team & NeuroAI Lab | BRAIN — Closed-Loop EEG Decoding — *Feb. 2026 – Jun. 2026*

- Built a **real-time EEG processing and decoding pipeline** for closed-loop BCI applications.
- Implemented **feature extraction, classification, and temporal smoothing** for online neural-state decoding.
- Developed **cross-subject adaptation** strategies to improve decoding robustness.

Hummel Lab, EPFL | Research Intern in Clinical Neuroengineering — *Jul. 2025 – Sep. 2025*

- Developed a complete **PsychoPy behavioral experiment** for clinical neuroscience research.
- Migrated an experimental paradigm from **MATLAB/Psychtoolbox to Python/PsychoPy**.
- Implemented **stimulus presentation, response acquisition, event synchronization, and data logging**.

EDUCATION

EPFL | MSc Neuro-X & EPFL AI Team member — *2025 – Present*
EPFL | BSc Microengineering, Bioengineering Track — *2021 – 2025*

SELECTED PROJECTS

CLoNeD — Closed-Loop Visual Neuromodulation

- Developed an **adaptive closed-loop optimization framework** robust to neural-state drift.
- Implemented **state-aware optimization** for targeted brain-region activation and multi-target tracking.

Neural Signal Processing — fMRI & EMG

- Built signal-processing pipelines for **fMRI and EMG**, including filtering, artifact removal, and feature extraction.
- Applied **time-series and spectral analysis** to physiological datasets.

TECHNICAL SKILLS AND LANGUAGES

Programming & Data: Python, MATLAB, C/C++, NumPy, SciPy, Pandas
Machine Learning: PyTorch, classification, feature extraction, model evaluation
Neural Signal Processing: EEG, EMG, fMRI, filtering, spectral analysis, artifact cancellation, time-series
BCI & Experimental Systems: Real-time signal processing, closed-loop systems, PsychoPy, Psychtoolbox
Embedded & Hardware: STM32, Arduino, KiCad, PCB design
French: Native · **English:** C1 / Advanced

LEADERSHIP & ACTIVITIES

Football Coach — Swiss 4th League | Led training and match preparation for a competitive senior team
Competitive Water Polo — Pays d'Aix Natation | French U15 Champion, 2018